

# Distilling Step-by-Step

Outperforming Larger Language Models with Less Training Data and Smaller Model Sizes

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# Outline

- 1 Motivation & Problem Definition
- 2 Methodology: Distilling Step-by-Step
- 3 Experimental Setup
- 4 Key Results
- 5 Ablation Studies Discussion
- 6 Conclusion

# 1. Motivation: The LLM Deployment Bottleneck

## The Power of LLMs:

- Models like PaLM (540B) and GPT-4 have emerged as reasoning agents.
- Excellent few-shot performance on complex tasks.

## The Practical Problem:

- **Size:** Serving a 175B+ model requires hundreds of GBs of VRAM[cite: 28].
- **Latency:** Inference is too slow for real-time applications.
- **Cost:** Running these models at scale is prohibitively expensive.

## The Goal

We need **small, task-specific models** (e.g., T5) that can match LLM performance without the massive compute cost.

# 1. Why Existing Solutions Fail

Standard approaches to training small models have significant downsides:

## A. Standard Finetuning

- Updates a small model (e.g., BERT/T5) using human labels.
- **Flaw:** Requires large amounts of **human-annotated data**, which is expensive and slow to collect [cite: 33-35].

## B. Standard Distillation

- Uses an LLM (Teacher) to generate pseudo-labels for unlabeled data.
- **Flaw:** Requires massive amounts of **unlabeled data** to bridge the performance gap[cite: 36].

## 2. The Core Idea: Rationales

- **Paradigm Shift:** Don't just view LLMs as label generators. View them as *reasoners*[cite: 39].
- **Rationales:** Chain-of-Thought (CoT) explanations that justify a prediction.
- **Why it helps:** A label provides 1 bit of information (Positive/Negative). A rationale provides the *logic* behind the decision.

### Example (Math Problem)

**Input:** "Jesse's room is 11x15. She has 16 sq ft of carpet. How much more does she need?"

**Rationale:** "Area = length \* width.  $11 * 15 = 165$ .  $165 - 16 = 149$ ."

**Label:** "149"

## 2.1 Methodology: Extracting Rationales (CoT)

How do we get these rationales?

- 1 Take an unlabeled dataset  $D = \{x_i\}$ .
- 2 Curate a **Few-Shot Prompt**  $p$  containing triplets:  $(x^p, r^p, y^p)$  [cite: 127-128].
  - Input ( $x$ ), Rationale ( $r$ ), Label ( $y$ ).
- 3 Feed  $x_i$  to the Teacher LLM (PaLM 540B).
- 4 The Teacher mimics the reasoning to generate a rationale  $\hat{r}_i$  and label  $\hat{y}_i$ .

*This creates a synthetic, rationale-rich dataset.*

## 2.2 Methodology: Multi-Task Training

We train the Student Model  $f$  (e.g., T5) on **two tasks** simultaneously using task prefixes[cite: 162]:

### Task 1: Label Prediction

- **Input:** [label] + Input Text
- **Target:** Predicted Label ( $\hat{y}$ )

### Task 2: Rationale Generation

- **Input:** [rationale] + Input Text
- **Target:** Explanatory Rationale ( $\hat{r}$ )

**Crucial Deployment Detail:** At test time, we **only** run Task 1. We discard the rationale head, so there is **no computational overhead**[cite: 161].

## 2.3 Mathematical Formulation

The training objective minimizes a weighted sum of two losses[cite: 155]:

$$\mathcal{L} = \mathcal{L}_{label} + \lambda \mathcal{L}_{rationale} \quad (1)$$

**1. Label Loss ( $\mathcal{L}_{label}$ ):** Standard Cross-Entropy loss between prediction and target label.

$$\mathcal{L}_{label} = \frac{1}{N} \sum_{i=1}^N \ell(f(x_i), \hat{y}_i)$$

**2. Rationale Loss ( $\mathcal{L}_{rationale}$ ):** Loss for generating the correct explanation tokens.

$$\mathcal{L}_{rationale} = \frac{1}{N} \sum_{i=1}^N \ell(f(x_i), \hat{r}_i)$$

### 3. Experimental Setup: Datasets

Tested on 4 benchmark NLP datasets covering 3 distinct tasks [cite: 173-175]:

| Dataset       | Task Type                  | Challenge                            |
|---------------|----------------------------|--------------------------------------|
| <b>e-SNLI</b> | Natural Language Inference | Logical entailment with explanations |
| <b>ANLI</b>   | Adversarial NLI            | Hard examples that fool models       |
| <b>CQA</b>    | Commonsense QA             | Requires world knowledge             |
| <b>SVAMP</b>  | Math Word Problems         | Arithmetic reasoning                 |

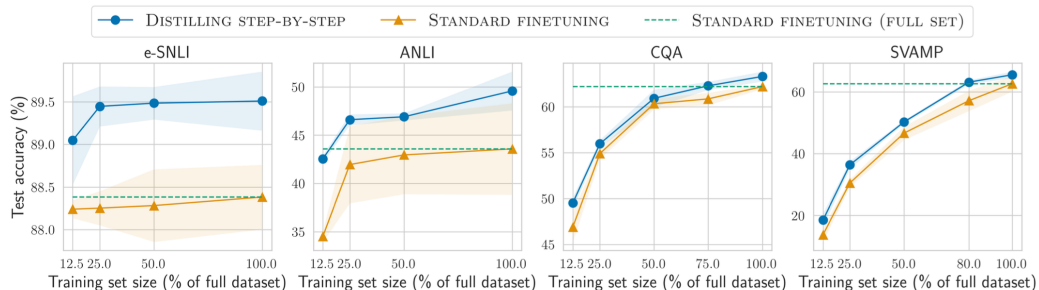
#### Models Comparison:

- **Teacher:** PaLM (540 Billion Parameters)
- **Student:** T5-Base (220M), T5-Large (770M), T5-XXL (11B)

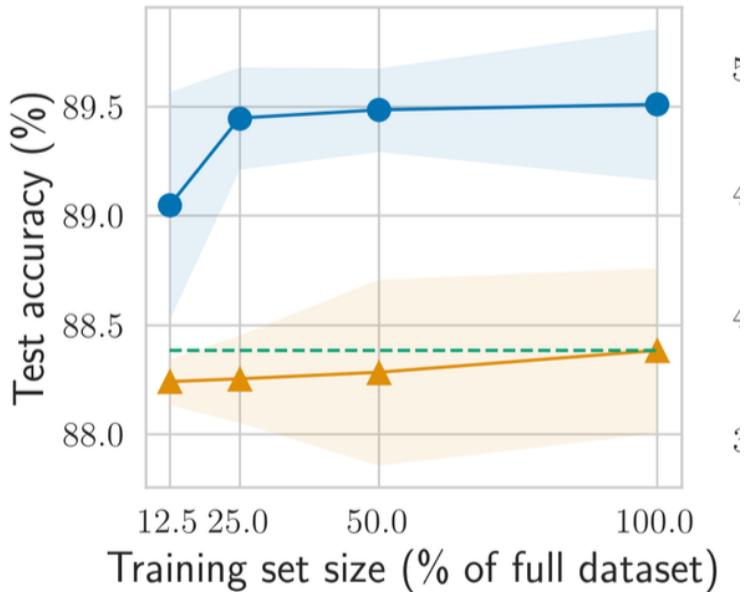
## 4.1 Result: Data Efficiency (Labeled Data)

*Can we match standard finetuning with less data?*

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## e-SNLI



## 4.1 Result: Data Efficiency (Labeled Data)

*Can we match standard finetuning with less data?* **Findings (Figure 4) [cite: 185-186]:**

- On **e-SNLI**, Distilling Step-by-Step with only **12.5%** of the data outperforms Standard Finetuning with **100%** of the data.
- On **ANLI**, we achieve the same performance with **75%** less data.

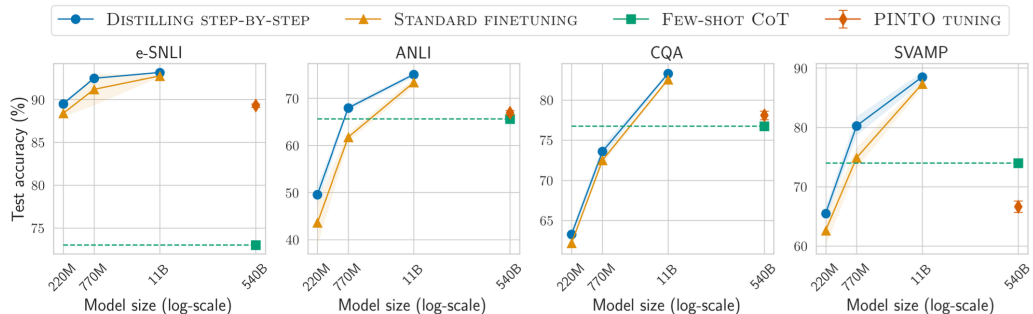
### Implication

High-quality rationales compensate for small dataset sizes.

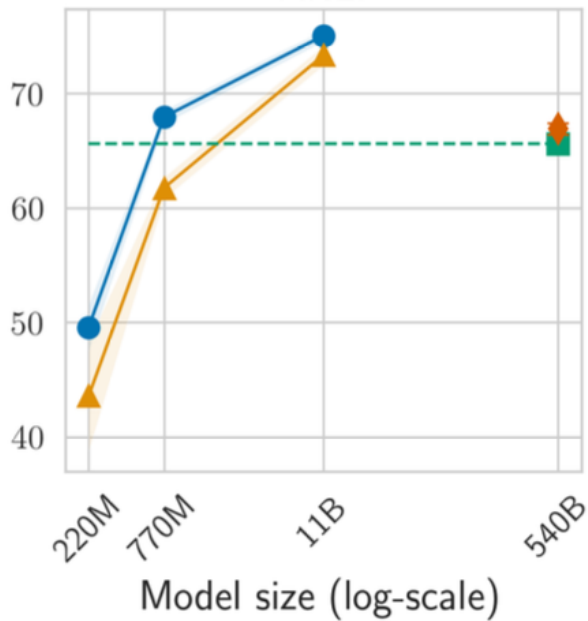
## 4.2 Result: Model Size Efficiency

*Can a small student beat the giant teacher?*

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# ANLI



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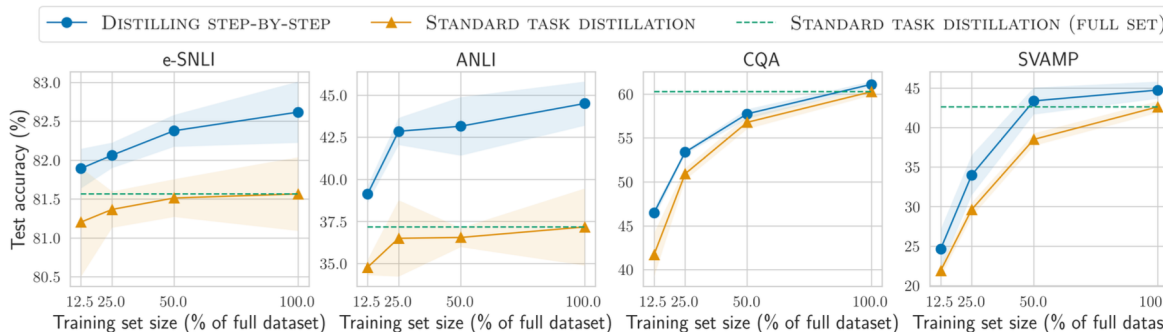
**Findings (Figure 6) [cite: 283-286]:**

- A **770M T5** model (Distilled) outperforms the **540B PaLM** (Few-Shot CoT) on ANLI.
- **Compression Ratio:** The student is over **700x smaller**.
- On e-SNLI, even the tiny **220M model** beats the 540B teacher.

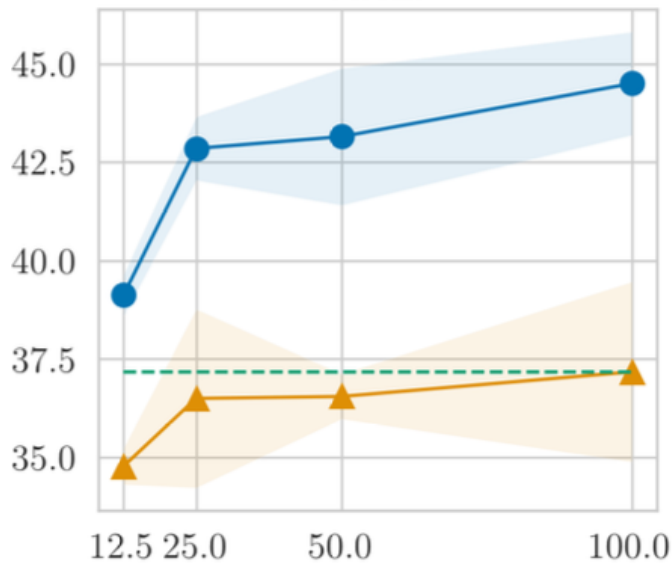
This enables deploying "LLM-level intelligence" on a single standard GPU or even CPU.

## 4.3 Result: Unlabeled Data Efficiency

*What if we have NO human labels?*



# ANLI



) Training set size (% of full dataset)

## 4.3 Result: Unlabeled Data Efficiency

*What if we have NO human labels?*

We compared against **Standard Task Distillation** (using only Teacher labels, no rationales) [cite: 190-192].

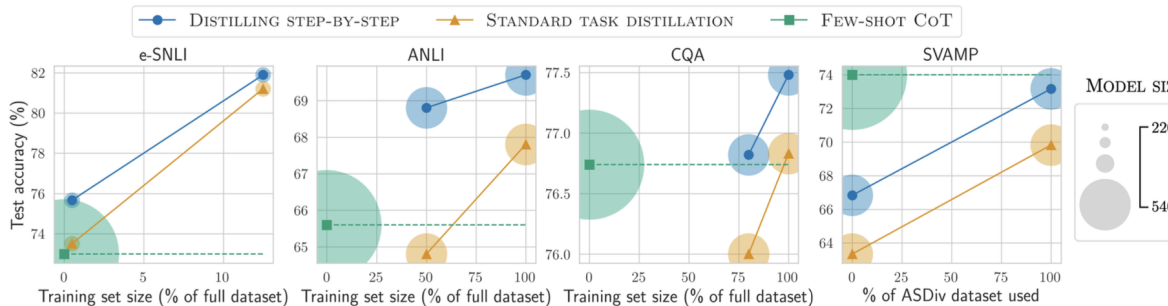
- **Finding:** Distilling Step-by-Step consistently outperforms Standard Distillation.
- **Example:** On ANLI, we match the teacher's performance using only **12.5%** of the unlabeled data, whereas standard distillation needs 100%.

## 4.4 Result: Data Augmentation

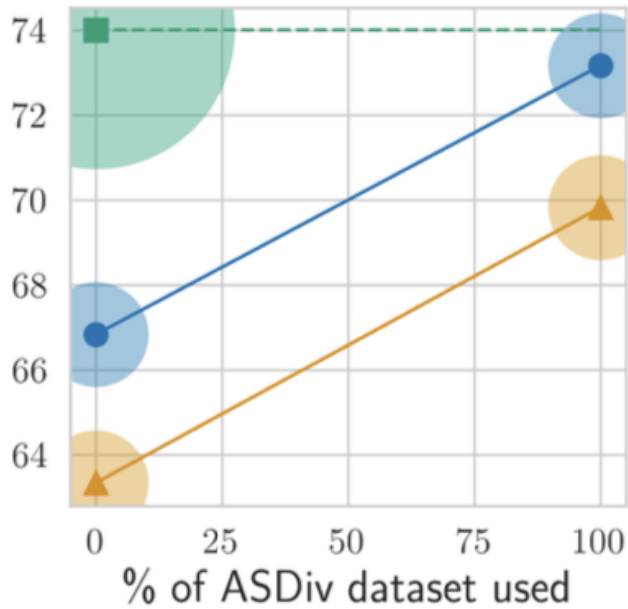
On small datasets like SVAMP (Math), performance gaps can remain.

### Experiment:

- Augmented SVAMP with unlabeled data from ASDiv[cite: 388].
- **Result:** Distilling Step-by-Step exploited the extra unlabeled data much better than standard distillation.
- The student model (11B) eventually matched the 540B Teacher using the augmented data (Figure 9).



# SVAMP



## 5.1 Ablation: Why Multi-Task?

Why not just concatenate the rationale to the input? Or output `[rationale, label]` as one sequence?

| Method               | Dataset      |              |              |              |
|----------------------|--------------|--------------|--------------|--------------|
|                      | e-SNLI       | ANLI         | CQA          | SVAMP        |
| STANDARD FINETUNING  | 88.38        | 43.58        | 62.19        | 62.63        |
| SINGLE-TASK TRAINING | 88.88        | 43.50        | 61.37        | 63.00        |
| MULTI-TASK TRAINING  | <b>89.51</b> | <b>49.58</b> | <b>63.29</b> | <b>65.50</b> |

## 5.1 Ablation: Why Multi-Task?

Why not just concatenate the rationale to the input? Or output [rationale, label] as one sequence?

**Comparison (Table 2) [cite: 511-512]:**

- **Single-Task Training:** Treats rationale and label as one sequence.
- **Outcome:** Often performs *worse* than standard finetuning.
- **Reason:** The generation of long rationales can interfere with the label prediction if not separated into distinct tasks.

**Conclusion:** The Multi-Task formulation (Equation 3) is essential for success.

## 5.2 Ablation: Does Teacher Size Matter?

Does this only work with massive Google-scale models like PaLM 540B?

| Method                  | LLM  | Dataset |       |       |       |
|-------------------------|------|---------|-------|-------|-------|
|                         |      | e-SNLI  | ANLI  | CQA   | SVAMP |
| STANDARD FINETUNING     | N/A  | 88.38   | 43.58 | 62.19 | 62.63 |
| DISTILLING STEP-BY-STEP | 20B  | 89.12   | 48.15 | 63.25 | 63.00 |
| DISTILLING STEP-BY-STEP | 540B | 89.51   | 49.58 | 63.29 | 65.50 |

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### Experiment:

- Tested using **GPT-NeoX (20B)** as the teacher instead of PaLM (540B)[cite: 501].

### Result (Table 1):

- Distilling Step-by-Step still provides performance gains over standard finetuning.
- However, gains are smaller compared to using PaLM.
- *Lesson: Better Teachers provide better Rationales, leading to better Students.*

## 5.3 Critical Analysis

### Strengths:

- ① **Resource Efficiency:** Reduces data needs and model size simultaneously.
- ② **Deployability:** Decouples "thinking" (training) from "inference" (deployment).
- ③ **Versatility:** Works across classification, NLI, and Math tasks.

### Limitations [cite: 523-528]:

- ① **Prompt Dependency:** Requires carefully designed few-shot prompts to extract good rationales.
- ② **Hallucinations:** If the Teacher hallucinates a wrong rationale, the Student learns it.
- ③ **Complexity Limit:** LLMs struggle with complex planning; this method cannot distill capabilities the Teacher doesn't have.

- ❶ **Rationales are Rich Supervision:** Step-by-Step distillation extracts more "bits" of intelligence per example than simple labels.
- ❷ **Less Data Required:** We can beat state-of-the-art baselines with 50-75% less training data.
- ❸ **Tiny Giants:** We can deploy 770M parameter models that outperform 540B parameter models.

## Final Thought:

*We don't need giant models for every user query—we need giant models to teach small models how to reason.*

# Thank You!